

# Electroplating: Copper Coating a Key

Discovering how electricity can transfer a shiny layer of copper metal onto an ordinary iron key!

## 1. Introduction: Why is One Key Dull and the Other Shiny?

Have you ever noticed how an old iron key looks dark, rusty, and dull, while brand-new brass keys or ornaments sparkle with a bright metallic shine? Through the power of electricity, we can take an ordinary dull iron key and coat it with a thin, gleaming layer of copper metal!

## Materials Required

Copper Sulphate  
Solution

Copper Strip

Iron Key

9V Battery

Connecting  
Wires

## 2. What is Electroplating?

**Electroplating** is the process of depositing a thin layer of a desired metal onto another object using an electric current.

| Component                | Role in Experiment             | Specific Item Used                                                      |
|--------------------------|--------------------------------|-------------------------------------------------------------------------|
| Anode (+)                | The Copper "Giver"             | Copper Metal Strip                                                      |
| Cathode (-)              | The Key "Receiver"             | Cleaned Iron Key                                                        |
| Electrolyte Solution     | Medium for Metal Ion Transport | Blue Copper Sulphate Solution ( <i>Neela Thotha</i> / $\text{CuSO}_4$ ) |
| Power Source & Conductor | Drives the Electric Current    | 9V Battery & Connecting Wires                                           |



### Safety First! Lab & Experiment Tips

- **Protective Gear:** Always wear safety goggles and gloves to protect your eyes and hands.
- **Low Voltage Only:** Use only a small 9V battery with connecting wires. NEVER connect experiments to wall power sockets!
- **Chemical Care:** Handle Copper Sulphate ( $\text{CuSO}_4$ ) carefully — do not taste, touch directly, or ingest.
- **Clean Workspace:** Keep your experiment area completely clean, tidy, and dry at all times.

## 3. How to Copper Coat a Key: Step-by-Step Procedure

- 1 Clean the Iron Key:** Thoroughly scrub the iron key with sandpaper or vinegar to remove rust, oil, and dirt.
- 2 Prepare the Solution:** Pour the blue copper sulphate solution into a glass container or beaker.

3

**Connect Anode Wire:** Attach the copper metal strip to the positive terminal (+) wire of the 9V battery using a connecting wire.

4

**Connect Cathode Wire:** Attach the cleaned iron key to the negative terminal (-) wire of the 9V battery using another connecting wire.

5

**Submerge in Solution:** Dip both the copper strip and the iron key into the blue copper sulphate solution, ensuring they do not touch each other directly.

6

**Wait and Watch:** Allow electric current to flow through the wires for 10–15 minutes. Watch a reddish-brown shiny copper layer form on the iron key!

## 4. Real-Life Applications of Electroplating

### 🌟 Temple Ornaments & Puja Items

Gold-plated temple kalash, oil lamps (diyas), and decorative idols.

### 🔍 Utensils & Hardware

Brass-coated cooking utensils, shiny door handles, and metal locks.

### 🚲 Vehicle & Household Fittings

Chrome-plated bicycle handlebars, car bumpers, and shiny bathroom taps.

### 💍 Artificial Jewellery

Faux gold/silver-coated necklaces, bangles, and copper-plated accessories.

## STUDENT WORKSHEET & ASSESSMENT

Name: \_\_\_\_\_

Date: \_\_\_\_\_

### Section A: Materials & Equipment Identification

**Q1. List the 5 essential materials required to set up the copper electroplating experiment:**

- |          |          |
|----------|----------|
| 1) _____ | 2) _____ |
| 3) _____ | 4) _____ |
| 5) _____ |          |

### Section B: Multiple Choice Questions

**Q2. Which part of the electroplating setup gives copper to the key?**

- (a) The 9V battery
- (b) The copper strip (Anode)
- (c) The iron key (Cathode)
- (d) The glass beaker

**Q3. To electroplate an iron key with copper, the iron key must be connected via wires to which terminal of the battery?**

- (a) Positive (+) terminal
- (b) Negative (-) terminal
- (c) Neutral terminal
- (d) Either positive or negative terminal

**Q4. What is the common Indian name for blue copper sulphate solution used as the electrolyte?**

- (a) Baking Soda
- (b) Neela Thotha
- (c) Caustic Soda
- (d) Fitkari

**Q5. Why is a 9V battery recommended for this experiment instead of connecting wires directly to a wall power outlet?**

- (a) 9V battery current is blue in colour
- (b) Wall electricity is low voltage and ineffective
- (c) Wall electricity carries dangerously high voltage and can cause fatal electric shocks
- (d) Batteries produce faster copper melting

### Section C: Short Answer & Thought-Provoking Questions

**Q6. Fill in the blanks with appropriate scientific terms:**

- a) The electrode connected via wire to the positive terminal of the 9V battery is called the \_\_\_\_\_.
- b) The electrode connected via wire to the negative terminal of the 9V battery is called the \_\_\_\_\_.
- c) In this experiment, the liquid electrolyte solution used is \_\_\_\_\_ solution.

**Q7. Why do you think electroplating an iron key with copper using electric current and wires is better than simply painting it with copper-coloured paint?**

**Q8. List two real-life examples where electroplating is used to make objects look attractive or protect them from rusting.**

**Q9. Suppose a student accidentally connects the iron key to the positive (+) terminal wire and the copper strip to the negative (-) terminal wire. What would happen during the experiment?**